Nutrient Depletion Stimulates AMPK to Inhibit mTORC1 and Trigger Protective Autophagy

Authors: Julian Rivera, PhD, Hannah Zhao, PhD
DOI: 10.1016/j.cmet.2024.02.005 | PMID: 38675005 | PMCID: PMC10875005

Page 1 — Abstract

Metabolic homeostasis requires precise coordination of energy-sensing kinases during nutritional starvation.

Page 2 — Introduction

Depletion of cellular ATP increases the AMP/ATP ratio, directly activating AMPK by liver kinase B1 phosphorylation.

Page 3 — Methods

Active AMPK phosphorylates the tuberous sclerosis complex subunit TSC2, enhancing its GTPase-activating activity toward Rheb.

Page 4 — Results: Kinase Inhibition

Inactivation of Rheb downregulates mTORC1 kinase activity, suppressing ribosomal protein S6 kinase phosphorylation.

Page 5 — Results: Autophagy Induction

Relief of mTORC1 suppression activates ULK1 kinase, which phosphorylates Beclin-1 to stimulate autophagosome formation.

Page 6 — Discussion

Together, these data establish that AMPK suppresses mTORC1 to stimulate autophagy under energy stress. Inactive mutants did not stimulate autophagosome formation.